I4 Numerical methods in context A level only

Name: ____________________

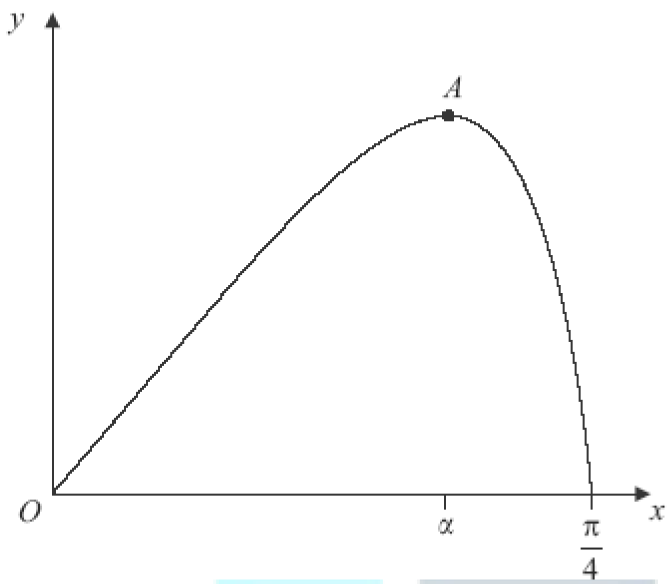
Class: ____________________

Date: ____________________

Time:	**17 minutes**
Marks:	**14 marks**
Comments:	

Q1.

A curve is defined for $0 \le x \le \frac{\pi}{4}$ by the equation $y = x \cos 2x$, and is sketched below.



(a) Find $\frac{dy}{dx}$.

(2)

(b) The point A, where $x = \alpha$, on the curve is a stationary point.

(i) Show that $1 - 2\alpha \tan 2\alpha = 0$.

(2)

(ii) Show that $0.4 < \alpha < 0.5$.

(2)

(iii) Show that the equation $1 - 2x \tan 2x = 0$ can be rearranged to become $x = \frac{1}{2}\tan^{-1}\left(\frac{1}{2x}\right)$.

(1)

(iv) Use the iteration $x_{n+1} = \frac{1}{2}\tan^{-1}\left(\frac{1}{2x_n}\right)$ with $x_1 = 0.4$ to find x_3, giving your answer to two significant figures.

(2)

(c) Use integration by parts to find $\int_0^{0.5} x \cos 2x \, dx$, giving your answer to three significant figures.

(5)

(Total 14 marks)